Priyanka Chatterjee

Senior Software Engineer (Backend/Architect) · Python/Go/Node.js · Agentic AI & RAG · AWS · 9,300★ OSS Contributor · 3.5+ Years

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SUMMARY

Senior Software Engineer (Backend/Architect) with 3.5+ years of experience and expertise in **Python**, **Go**, **Node.js**, **PostgreSQL**/**MongoDB**, and **event-driven** microservices on **AWS**. Built a production **RAG** agent and multi-agent AI system at Turbot HQ — a product company, not services — and led an **AWS** SDK migration across 180+ call sites and 11+ **AWS** services, cutting resource-discovery latency 70-90%.

TECHNICAL SKILLS

Languages: Python, Go, Node.js, TypeScript, JavaScript (ES6+), SQL, Bash

AI & Agentic Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI, Claude Code, Cursor, Codex

Backend & APIs: REST, gRPC, Microservices, Event-Driven Architecture, OAuth 2.0 (Auth0), JWT

Databases: PostgreSQL, MySQL, MongoDB, DuckDB; jsonb, CTEs, query tuning, schema design

Cloud & Containers: AWS (Lambda, EC2, S3, SecurityHub, GuardDuty), Azure, GCP, Docker, Kubernetes/k8s (working knowledge; collaborate with platform teams)

DevOps & Testing: GitHub Actions, CI/CD, Git, Unit Testing, Integration Testing, Code Review

WORK EXPERIENCE

Turbot HQ, Inc. Remote — *August 2022 – April 2026*

Software Engineer I

- Built the Steampipe Model Context Protocol (MCP) server end to end in **Python** — a 0→1 build, from identifying the gap to shipping a production capability — giving **LLM** agents standardized, real-time access to enterprise cloud and security data.
- Designed a multi-agent AI system on the OpenAI Agents SDK — a production **agentic**-orchestration framework — orchestrating **LLM** agents that generate policy commands, validate them in a live environment, self-correct failures, and score outcomes against an internal metric.
- Built a **RAG**-based Pydantic AI agent with an embeddings-backed knowledge base in **AWS** S3 vector storage; deployed to **AWS** Lambda and integrated with Slack for instant, context-aware query resolution.
- Authored an **LLM**-driven Claude Code skill automating code refactoring, unit test generation, deployment, and PR creation — using AI as leverage in daily engineering; used it to migrate 35+ Azure cloud-governance modules to **event-driven** eventing, cutting resource-discovery latency 70-90%.
- Built a Kafka-based, **event-driven** distributed-systems pipeline where consumers asynchronously processed policy-enforcement events; added a dead-letter queue (DLQ) for reliable failure handling.
- Led the **AWS** SDK for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services in production **Node.js** Lambdas; reduced policy-code duplication ~60% via templating, keeping the platform supported past **AWS** SDK v2 EOL.
- Designed and developed an internal **REST** API microservice in **Go** using the Gin framework as part of a greenfield build of Turbot Guardrails, an enterprise B2B SaaS platform; containerized with **Docker**, deployed to **AWS** EC2, and served as a gRPC client for several internal **event-driven** microservices.
- Optimized complex **PostgreSQL** queries powering customer compliance audits — rewriting **AWS** IAM policy-evaluation queries with jsonb array elements and multi-stage CTEs — improving query performance 40% and cutting enterprise audit time 50%, the same accuracy-and-audit-trail rigor a deterministic reconciliation engine demands.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN — *August 2019 – May 2023*

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design